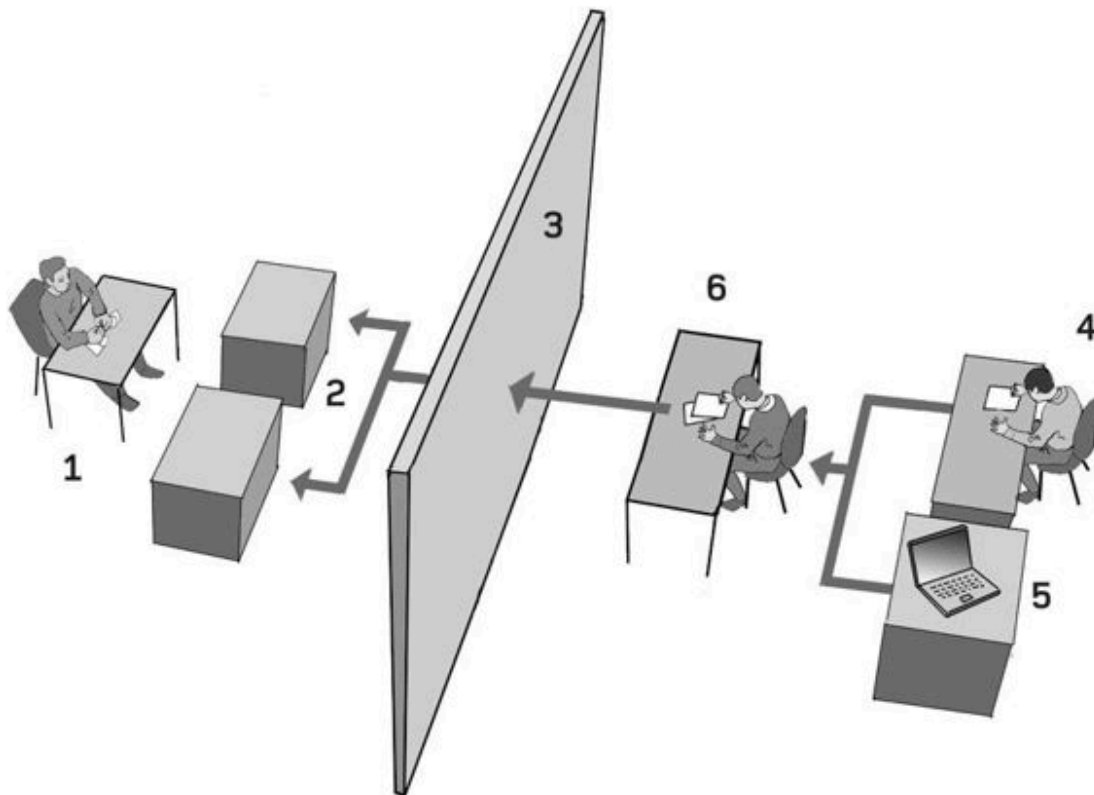


Artificial intelligence

Advances in science and technology have produced machines capable of performing tasks that previously could only be done by human thought. Sophisticated computer programming has taken them beyond the realm of simple calculation, and given them 'artificial intelligence', including the ability to use language. So machines can mimic human behaviour – but can they actually think? Mathematician and computer pioneer Alan Turing proposed a simple test of a machine's ability to show intelligence. A machine and a human are both asked questions in normal human language, to which they give replies in the same language. If an impartial judge cannot tell which answers are which, the machine has passed the test and is considered to be showing intelligence. Modern computers are increasingly able to do this successfully, especially as we programme them to process information in the same way as human brains and even introduce such concepts as 'fuzzy logic'. But can we therefore say that they are capable of 'thinking'? Will they ever be capable of what we understand to be consciousness?



Experimental set-up for a Turing test

1 Human questioner

2 Terminals to display responses

3 Barrier

4 Human answerer

5 Computer

6 Experiment controller relays either human or computer response